MIL-L-16991C
29 November 1973
SUPERSEDING
MIL-L-16991B
6 April 1966

MILITARY SPECIFICATION

LENS, OPHTHALMIC, GLASS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers glass ophthalmic lenses (see Section 6).

1.2 Classification. Lenses shall be of the following types, classes, styles, sizes and power required, as specified (see 6.3):

Type

I. Lens blanks.
II. Lens, single vision, uncut.
III. Lens, single vision, uncut, (minimum safety thickness).

Class

A. Bifocal, semifinished.
B. Molded, rough.
C. Semicylinder.
D. Semispheres.
E. Simple.
F. Compound.

Style

1. Spheres.
2. Plus plano cylinders.
3. Plus on plus compounds.
4. Minus on plus compounds.

FSC 6540

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Size. The following sizes reflect the lengths of the shortest and longest axis (see Figure 1).

a. 54 mm.
b. 54 mm x 57 mm.
c. 55 mm x 60 mm.
d. 57 mm x 59 mm.
e. 58 mm.
f. 60 mm.
g. 60 mm x 62 mm.
h. 62 mm.
i. Conforming to Figure 2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification, to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INCORPORATED

ANSI Z80.1 - 1972 American National Standard Requirements for First-Quality Prescription Ophthalmic Lenses.

(Application for copies should be addressed to the American National Standards Institute, Incorporated, 1430 Broadway, New York, N.Y. 10018.

3. REQUIREMENTS

3.1 Material.

3.1.1 Lens. The lens shall be made of high grade white ophthalmic quality crown glass, with an index of refraction of 1.5230, ± 0.0015. The glass shall have a high degree of clearness and transparency, and shall remain clear and not discolor when exposed to ordinary sunlight. It shall be durable and of suitable composition and hardness to give good service in use, resist the action of weather, and not be easily scratched or damaged under ordinary conditions. The glass shall be free of bubbles, inclusions, striae and cloudiness (See 4.3.1).

3.1.2 Segment. Where applicable the segment of the lens shall be made of glass of the type commonly known in the optical industry as barium. It shall be free of bubbles, inclusions, striae, and cloudiness. (See 4.3.1). The dispersion of the segment glass, Nu Value (v), shall be equal to, or greater than, the value defined by the formula:

$$v \text{ equals } \frac{16.7\ (n-1)}{n - 1.3744}$$

Where "n" is the index of refraction of the segment glass for the sodium D line.

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3.2 Construction.

3.2.1 Type I. Lens blanks.

3.2.1.1 Class A. Bifocal, semifinished.

3.2.1.1.1 Major blank with fused segment. The semifinished lens blank shall be fabricated of the material specified in 3.1.1 and shall have a fused straight top bifocal (see 3.2.1.1.2). The blank shall be free of bubbles or other defects which might result from the fusing process. The bifocal segment shall be of the addition specified, within a tolerance of plus or minus 0.09 diopter (see 6.2.4 for definition). The addition power shall be measured as specified in 4.4.1. The distance from the geometric center of the major blank to the top of the segment shall be 3.25 mm to 6 mm. Unless otherwise specified, the segment shall be centered, or decentered to the right or left 3.0 mm, plus 2.2, mm minus 1.0 mm. The segment side of the blank shall be ground and polished to the spherical base curve specified (see 6.3), within a tolerance of plus or minus 0.08 diopters and shall be in the corrective curve form. The front surface shall be free from any localized refractive error, due to any defect which impairs the image quality, when the questioned area is examined either visually or with a standard lens measuring instrument (see 4.4.1).

3.2.1.1.1.1 Thickness. Each class A blank shall be sufficiently thick to produce the lens for which it is designated on the manufacturers grinding chart with a final finished center thickness of 2.0 mm, minimum.

3.2.1.1.2 Segment. The segment shall be of the material specified in 3.1.2 and shall be of the straight top type. The segment diameter (measured at the widest portion) shall be 25 mm, ± 0.5 mm. When the lens blanks are paired, the paired segment diameters shall be equal within a tolerance of 0.5 mm. The optical center of the segment shall be 5 mm ± 0.5 mm below the flat top.

3.2.1.1.3 Base curves. The front surface curve supplied shall be such that when used in accordance with the corrected curve surfacing chart published by the manufacturer of the lens blank, the finished lens shall be in the corrected curve form (see 6.2.3) and to specified axial back vertex power, within a tolerance of plus or minus 0.06 diopters. The lens blanks shall be such that the finished prescription desired can be completed with tools of even multiples of 0.25 diopters only (to provide finished lens powers in 0.25 diopter increments).

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3.2.1.1.4 Coating of line. The top line of the flat top segment shall be coated to minimize reflections.

3.2.1.2 Class B. Molded rough. The rough molded lens shall be molded in meniscus form from the material specified in 3.1.1, with a curve of approximately 6 diopters on both sides.

3.2.1.2.1 Thickness. The thickness of class B lens at the center and edges shall be as specified.

3.2.1.3 Class C. Semicylinder. The semicylinder lens shall be molded in meniscus form from the material specified in 3.1.1. The convex surface shall be ground and polished toric, with a spherical base curve, as specified, in the corrected curve series, and a cylinder curve as specified (6.3). The concave surface shall be a rough, fine ground or molded surface with a curve of approximately 6 diopters.

3.2.1.3.1 Thickness. The thickness of the Class C lens at the center shall be sufficient to produce the designated lens on the manufacturer's grinding chart with final specified minimum of 2.0 mm.

3.2.1.4 Class D. Semisphere. The semisphere lens shall be molded in meniscus form from the material specified in 3.1.1. The plus base curve lens shall be ground and polished on the convex side. The minus base curve lens shall be ground and polished on the concave side. The applicable surface shall be ground and polished to the spherical base curve as specified (see 6.3), in the corrected curve series. The opposite surface shall be a rough, fine ground, or molded surface with a curve of approximately 6 diopters.

3.2.1.4.1 Thickness. The thickness of the Class D lens at the center shall be as specified in 3.2.1.3.1.

3.2.2 Type II. Lens, single vision, uncut.

3.2.2.1 Class E. Simple.

3.2.2.1.1 Style 1: Sphere. The lens shall be made of the glass specified in 3.1.1. The sphere shall be a single vision, uncut, simple lens with plus or minus spherical power. The axial power shall correspond with plus or minus spherical dioptric power specified, with allowable tolerances (3.2.2.3.1). The marginal power shall correspond with axial spherical and cylindrical power as specified, with allowable tolerances (3.2.2.3.2).

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3.2.2.2 Class F. Compound. The lens shall be made of the glass specified in 3.1.1.

3.2.2.2.1 Style 2. plus plano cylinders. The single vision, uncut lens shall be plano cylinder with the cylinder curve located on the convex surface. The axial power shall correspond with plus cylindrical power specified, with allowable tolerances (3.2.2.3.1). The marginal power shall correspond with axial spherical and cylindrical power, as specified, with allowable tolerances (3.2.2.3.2).

3.2.2.2.2 Style 3. plus on plus compounds. The single vision, uncut, plus on plus compound lens shall have the cylinder curve located on the convex surface. Axial power shall correspond with plus spherical and plus cylindrical power as specified, with allowable tolerance (3.2.2.3.1). The marginal power shall correspond with axial and cylindrical power as specified, with allowable tolerances (3.2.2.3.2).

3.2.2.2.3 Style 4. Minus on plus compounds. The single vision, uncut, minus on plus compound lens shall have the cylinder curve located on the convex surface. The axial power shall correspond with minus spherical and plus cylindrical power as specified, with allowable tolerances. (3.2.2.3.1). The marginal power shall correspond with axial spherical and cylindrical power as specified, with allowable tolerances (3.2.2.3.2).

3.2.2.3 Tolerances.

3.2.2.3.1 Axial power. The refractive power measured through the optical center shall agree to the specified power, to within the following tolerances:

Powers zero through 6.00	± 0.06 diopters.
Powers 6.00 through 12.00	± 1 percent of power.
Powers above 12.00	± 0.12 diopters.

3.2.2.3.2 Marginal power. The tolerances for marginal power shall be as follows:

(1) The marginal meridional power referred to the vertex sphere at points A and B (see figure 1) shall agree to the specified power within the tolerances for meridional power shown in tables 2 through 6, inclusive, of American Standards Institute (ANSI) Standard Z80.1 - 1972.

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(2) The cylindrical power, referred to the vertex sphere at points A and B (see figure 3) shall agree to the specified power within the tolerances for cylindrical power shown in tables 2 through 6 of ANSI Standard Z80.1 - 1972. The marginal power shall be measured as specified in 4.4.3.2.

3.2.2.4 Thickness. The axial thickness of type II lenses shall correspond to manufacturers specifications obtained from surfacing charts published by the manufacturer, except that the minimum center thickness of the lens shall be 2.0 mm. Thickness at a distance of 1.5 mm from any edge shall be a minimum of 1.0 mm for all lenses.

3.2.3 Type III. Lens, single vision, uncut, (minimum safety thickness.) The Type III lens shall be fabricated of the crown glass specified in 3.1.1. The uncut lenses shall conform to the requirements for Type II lenses, except that the thickness shall be as specified in 3.2.3.1.

3.2.3.1 Thickness. The minimum thickness of Type III lenses shall be 3.0 mm, except in the case of strong plus power. Lenses of strong plus power shall have a minimum thickness of 2.5 mm at the distance of 26 mm from the geometrical center. Thickness shall be measured as specified in 4.4.4.

3.3 Surface characteristics. The polished surfaces of the lens shall be free from surface waves, scratches, pits, grayness or watermarks (See 4.3.1).

3.4 Shape. Unless otherwise specified, the lens shall be round, square with rounded curves, rectangular with rounded corners or oval. The lens shall be furnished in the size specified, within a tolerances of $\pm$ 0.5 mm.

3.5 Workmanship. Workmanship shall be first class throughout. Ophthalmic lenses shall be free from striae, bubbles, chips, or other internal defects which would impair their ophthalmic quality or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in table I. Unit of product for examination purposes shall be the lens.

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table I, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be the lens.

Table I. Sampling plan

For examination	Inspection level	AQL	
		Major	Minor
Surface power, thickness, segment size and location	S-2	1.0	
Lens size, surface defects and internal defects	II	1.0	2.5
For tests			
Refractive power and addition power	S-3	*	

*Acceptance number shall be zero.

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4.3 Examination. The lens shall be examined for defects including, but not limited to those listed in table II.

Table II. Classification of defects

Category	Defect
Major	
101	Surface power out of tolerance
102	Defects (Pits, grayness, scratches, bubbles, striae or inclusions - located over 3 mm from any edge)
103	Thickness not as specified
104	Segment size out of tolerance.
105	Segment location not as specified.
106	Wave distortion
107	Cylinder curve on concave side.
Minor	
201	Size below minimum
202	Cloudiness or haze
203	Jagged and sharp edges

4.3.1 Physical quality and appearance. Lenses shall be inspected against a dark background, in light from an open-shaded, 40 watt incandescent daylight lamp, with the lens 12 inches from the light source. No pits, scratches, grayness, watermarks, bubbles, striae, or inclusions shall be acceptable. Minute hairline scratches shall not be a cause for rejection. Waves found by visual inspection shall be passable if no deterioration in image quality is found when the localized area is examined with a standard lens measuring instrument.

4.3.1.1 Glass defects. Lenses shall be examined for glass defects and other irregularities affecting the optical qualities, by moving the lens in front of the aperture of a lens measuring instrument. Sudden jumps or blurring of the image will indicate a glass defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Bifocal addition power. Using a recognized standard lens measuring instrument which gives a measure of vertex power, the bifocal addition power shall be obtained using the following procedure:

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Place the segment side of the bifocal toward the light source, and take the difference between the distance and reading portion powers. The measurement through the distance portion must be made as far above the optical center of the distance portion as the measurement through the segment is below the optical center of the distance portion. The target images should be centered by the use of auxiliary prisms. The bifocal addition power shall meet the requirements specified in 3.2.1.1.1.

4.4.2 Methods of measurement.

4.4.2.1 Axial powers. The axial powers may be measured on any standard lens measuring instrument capable of yielding the required accuracy.

4.4.2.2 Marginal powers. The marginal powers may be measured on any standard lens measuring instrument that has been appropriately modified to yield the required accuracy referred to the vertex sphere.

One modification of a standard instrument that is satisfactory, is to add a hemispherical surface (see figure 3) to act as the seating surface against which the lens under test is held. The radius of the surface is 27 mm. This surface is so arranged that the edge of the standard opening is tangent to the surface of the sphere. A dot is marked on the sphere to locate a point 30 degrees from the axis. The optical center of the lens is pressed against this dot during the test. Suitable prisms are used to bring the image of the tangent within the field of view of the observing telescope. When the measurements are made in this manner, the observed values are the values referred to the vertex sphere and no corrections are necessary.

4.4.3 Surface power. The surface power shall be measured with an instrument such as a lens measure, a lens clock or Geneva gage.

4.4.4 Thickness measurement. Thickness shall be measured using a standard lens caliper or similar suitable measuring instrument.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Two lenses (1 pair), of the stock number specified, constitute one unit of issue.

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5.1.2 Packaging quantities. The number of units of issue indicated in table I shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table I. Packaging Quantities

Type	Class	Style	Lens Thickness (mm.)	Lens power	Type of Unit Package	No. of Units (Pairs) to be Packaged in each Intermediate Carton	No. of Units (Pairs) to be Packed in each Exterior Container
I	A	--	--	--	Carton	5	180
I	B	--	--	--	Envelope	5	180
I	C	--	4	--	Envelope	10	360
I	C	--	6	--	Envelope	5	180
I	D	--	4	--	Envelope	10	360
I	D	--	6	--	Envelope	5	180
II	E	1	--	0.00 to 2.50	Envelope	20	720
II	E	1	--	2.25 to 4.75	Envelope	15	540
II	E	1	--	4.50 to 8.00	Envelope	10	360
II	E	2	--	0.25 to 2.00	Envelope	20	720
II	E	2	--	1.75 to 5.25	Envelope	15	540
II	E	2	--	5.00 to 6.00	Envelope	10	360
II	F	3	--	0.25 to 2.25	Envelope	20	720
II	F	3	--	2.00 to 5.00	Envelope	15	540
II	F	3	--	4.75 to 8.00	Envelope	10	360
II	F	4	--	0.25 to 1.50	Envelope	20	720
II	F	4	--	1.25 to 5.25	Envelope	15	540
II	F	4	--	5.00 to 8.00	Envelope	10	360

NOTE: Quantities indicated in Intermediate Package Column and Exterior Container Column in Table I are mandatory, except when the number of units indicated will not fit the standard intermediate carton, a variation of -5 only, for classes E and F, is permitted. Quantities indicated in Exterior container Column will be adjusted accordingly to maintain the standard of 36 intermediate cartons per Exterior Container.

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit packaging. Type I, class A lens blanks shall be individually enclosed in tissue or other suitable material to protect the lens from damage, and further packaged in an individual carton of appropriate size and design. Closure of carton shall be adequate to prevent spilling of contents under normal handling. Two individual cartons (right and left) placed together constitute a matched pair. All other lens blanks and lenses shall be packaged in individual envelopes of appropriate size and design.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-676, except that commercial colors shall be acceptable. Carton shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of, or damage to, the contents. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; ot PPP-B-636, class weather-resistant. Special requirements as specified in Table II of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

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5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribtuion.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 31 PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements as specified in Table I of PPP-B-636 shall apply for fiberboard boxes.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual lens carton or envelope. Each individual carton shall be marked with base curve, addition of the segment, diameter of the segment, right or left eye, and name or registered trade-mark of the contractor. Each envelope shall be marked with power, type, center thickness, and name or registered trade-mark of the contractor.

5.3.2 Intermediate package. Each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ophthalmic lenses are intended for use in correcting defective vision.

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6.2 Definitions.

6.2.1 Vertex sphere. The vertex sphere (see figure 3) is a spherical surface with a radius of 27 mm, and tangent to the rear surface of the lens at its axial point.

6.2.2 Marginal points. The marginal points A and B (see figure 3). are chosen for measurement of marginal power. Points A and B lie on the axis of maximum and minimum power, respectively, and together with the optical center subtend angles of 30° at the center of the vertex sphere.

6.2.3 Corrected curve lens. A corrected curve lens shall be one member of an integrated series of ophthalmic lenses having surface curvatures especially computed and charted, which reduce the visually important monochromatic aberrations to the least significant feasible in commercial manufacture, for at least a 60 degree vista measured on the eye side, for specified centers of rotation and object distances.

6.2.4 Power of an addition. The power of an addition is defined as the difference between the powers measured through the distance and segment portions of the blank when the segment side is placed against the stop of a recognized standard lens measuring instrument.

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, style, size and power required (see 1.2).
(c) Base Curve and addition power required (if applicable).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.4 This specification does not include all types, classes, styles or sizes of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the items, which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

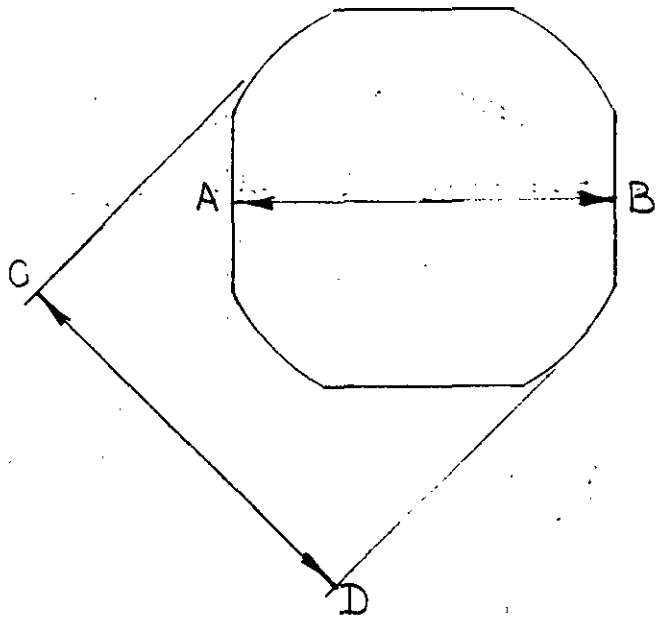
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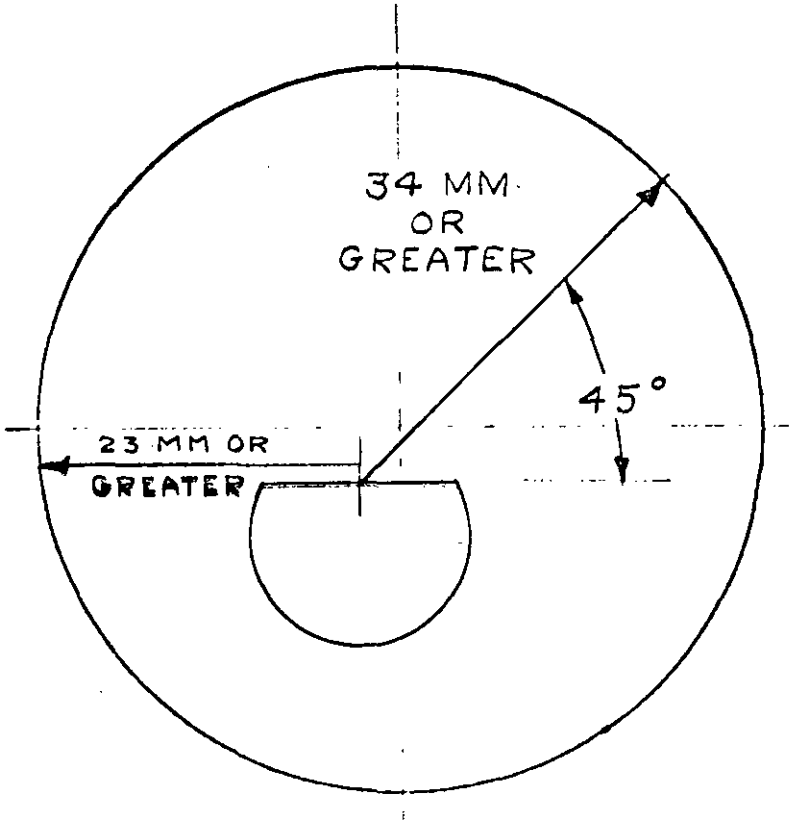


A - B - Shortest axis
C - D - Longest axis

An equivalent round lens is one with a diameter of C - D

FIGURE 1. Lens Measurements

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Additional Requirements:

Centered Segment - Minimum Length of Horizontal Axis - 58 MM.
Decentered Segment - Minimum Length of Horizontal Axis - 54 MM.

FIGURE 2 Size Requirements of
Type 1, Class A, Lens Blanks.

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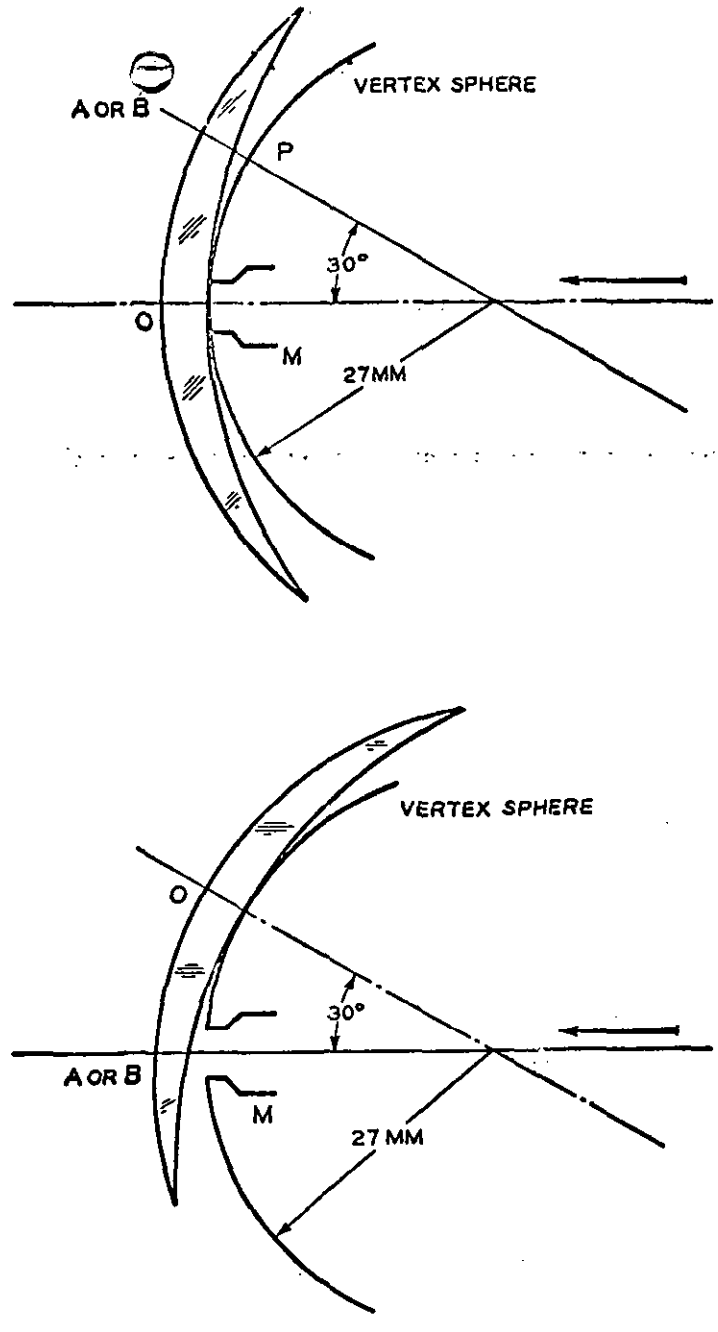


Fig. 3
Lens Testing on the Modified Vertometer

NOTE: Schematic diagram showing the positions assumed by a lens when tested on the modified vertometer. The upper part of the figure shows the arrangement for the measurement of axial power. The fixed reference opening against which the lens is pressed is shown at M. The optical center O is made to coincide with the center of M. The trace of the vertex sphere is also shown. In the lower part of the figure, point O is pressed against the vertex sphere as shown. Light from M passes through the lens under test at points A or B and the marginal power can thus be measured with respect to the vertex sphere.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-L-16991C

NAME OF ORGANIZATION AND ADDRESS

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MATERIAL PROCURED UNDER A
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